The documentation and process conversion measures necessary to comply with this revision shall be completed by 8 September 2001.

INCH-POUND

MIL-PRF-19500/548D
8 June 2001
SUPERSEDING
MIL-PRF-19500/548C
10 October 1997

PERFORMANCE SPECIFICATION

COUPLER, OPTO ELECTRONIC, SEMICONDUCTOR DEVICE, SOLID STATE
TYPES 4N47, 4N47A, 4N48, 4N48A, 4N49, 4N49A, 4N47U, 4N48U, AND 4N49U
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for solid state optically coupled isolators in which a gallium aluminum arsenide diode light source is optically coupled to a silicon NPN phototransistor. Four levels of product assurance are provided for each type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (4N47, 4N47A, 4N48, 4N48A, 4N49, and 4N49A)(similar to T0-99); see figure 2 (4N47U, 4N48U, and 4N49U).

1.3 Infrared-emitting diode maximum rating.

V_R	I_F (1) (2)	I_P (3)
V dc	mA dc	A (pk)
2	40	1

(1) Derate linearly to 125°C at 0.67 mA/°C above +65°C.
(2) Minimum recommended operating I_F is 1 mA dc at +25°C.
(3) 1.0 µs pulse width, 300 pps.

1.3.1 Phototransistor maximum rating.

V_{CEO}	V_{CBO}	V_{EBO}	$P_{T(1)}$	I_C
V dc	V dc	V dc	mW	mA dc
40	45	7	300	50

(1) Derate linearly to 125°C at 3 mW/°C above +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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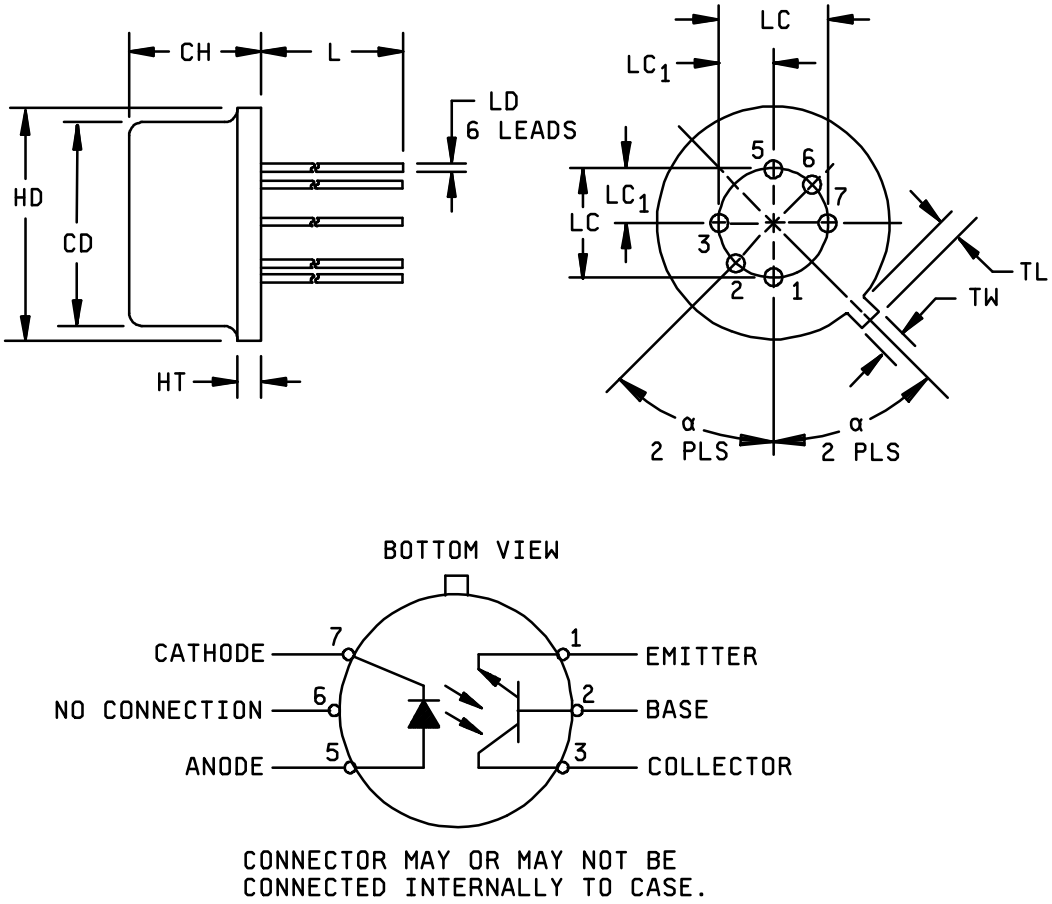


FIGURE 1. Dimensions and configuration (4N47, 4N47A, 4N48, 4N48A, 4N49, and 4N49A).

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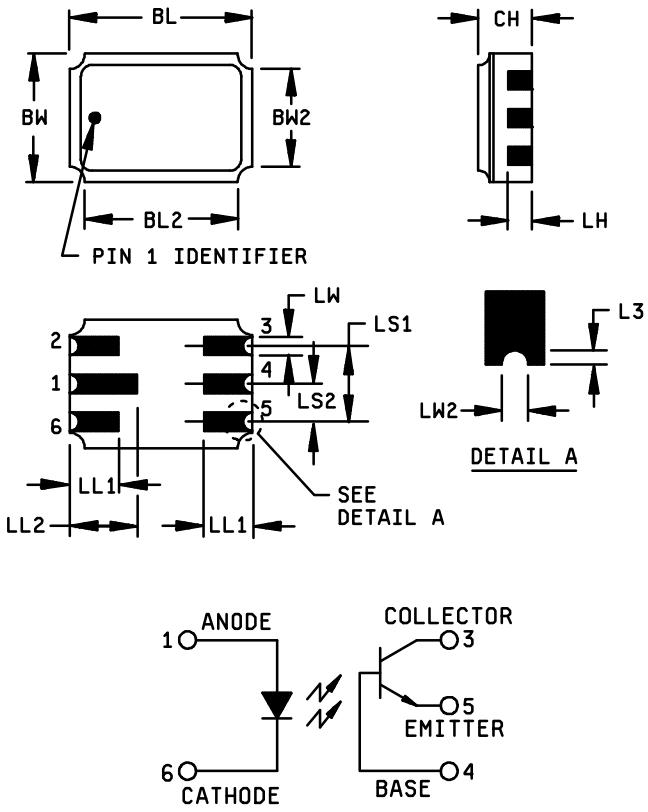
Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.155	.185	3.94	4.70	
HD	.335	.370	8.51	9.40	
HT		.040		1.02	
LC	.200 T.P.		5.08 T.P.		3
LC1	.100 T.P.		2.54 T.P.		3
LD	.016	.019	0.41	0.48	
LL	.500	.600	12.70	15.24	
TL	.029	.045	0.74	1.14	
TW	.028	.034	0.71	0.86	
α	45° T.P.		45° T.P.		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. T.P. designates true position. Leads having maximum diameter .019 (0.48 mm) measured in gauging plane .054 +.001, -.000 (1.37 +0.03, -0.00 mm) below the seating plane of the device shall be within .007 (0.18 mm) of their true position relative to a maximum width tab.
4. In accordance with ANSI Y14.5M, diameters are equivalent to Φx symbology.
5. Device types 4N47A, 4N48A, and 4N49A have the collector isolated from the case.

FIGURE 1. Dimensions and configuration (4N47, 4N47A, 4N48, 4N48A, 4N49, and 4N49A) - Continued.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.240	.250	6.10	6.35
BL2		.250		6.35
BW	.165	.175	4.19	4.44
BW2		.175		4.44
CH	.066	.080	1.68	2.03
LH	.026	.034	0.66	0.88
LL1	.060	.070	1.65	1.78
LL2	.082	.098	2.08	2.49
LL3	.003	.007	0.08	0.18
LS1	.095	.105	2.41	2.67
LS2	.045	.055	1.14	1.39
LW	.022	.028	0.56	0.71
LW2	.006	.022	0.15	0.56
N	6			
ND	3			
NE	0			



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension N: Number of terminals
4. Dimension ND/NE: Number of terminals per package side.

FIGURE 2. Dimensions and configuration (4N47U, 4N48U, and 4N49U).

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1.3.2 Total device ratings.

T_J	T_{STG}	V_{IO}
°C	°C	V dc
-65 to +125	-65 to +125	1,000 V (max)

1.4 Primary electrical characteristics at T_A = +25°C.

1.4.1 LED (input) characteristics at T_A = +25°C.

Limits	I_R V_R = 2 V dc	V_F I_F = 10 mA dc	I_F (1) V_F = 1.5 V dc
	µA dc	V dc	mA dc
Minimum		0.8	10
Maximum	100	1.5	

(1) Under no circumstances, shall I_F exceed the maximum rating of the device during the test.

1.4.2 Phototransistor (output) characteristics.

Limits	I_C off V_{CE} = 20 V	I_{CB} off V_{CB} = 20 V	$V_{(BR)CEO}$ I_C = 1 mA	$V_{(BR)CBO}$ I_E = 100 µA	$V_{(BR)EBO}$ I_E = 100 µA	h_{FE} V_{CE} = 5 V dc, I_C = 10 mA dc, I_F = 0
	nA	nA	V dc	V dc	V dc	
Minimum			40	45	7	100
Maximum	100	10				

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1.4.3 Coupled (transfer) characteristics.

Limits	R_{IO} at 1,000 V see 4.5.5	C_{IO} V = 0, f = 1 MHz see 4.5.5	Phototransistor mode switching (see figure 3)						Photodiode mode switching (see figure 3)	
			t_r			t_f			t_r	t_f
			4N47 4N47A 4N47U	4N48 4N48A 4N48U	4N49 4N49A 4N49U	4N47 4N47A 4N47U	4N48 4N48A 4N48U	4N49 4N49A 4N49U		
	Ohms	Pf	µs	µs	µs	µs	µs	µs	µs	µs
Minimum	10^{11}									
Maximum		5	20	20	25	20	20	25	3	3

Limits	$V_{CE(SAT)}$			$I_{C(ON)}$			$I_{CB(ON)}$
	4N47 4N47A 4N47U	4N48 4N48A 4N48U	4N49 4N49A 4N49U	4N47 4N47A 4N47U	4N48 4N48A 4N48U	4N49 4N49A 4N49U	
	I_F = 2 mA I_C = .5 mA	I_F = 2 mA I_C = 1 mA	I_F = 2 mA I_C = 2 mA	V_{CE} = 5 V I_F = 1 mA	V_{CE} = 5 V I_F = 1 mA	V_{CE} = 5 V I_F = 1 mA	V_{CB} = 5 V I_F = 10 mA
	V dc	V dc	V dc	mA dc	mA dc	mA dc	µA dc
Minimum				0.5	1.0	2.0	30.0
Maximum	0.3	0.3	0.3		5.0	10.0	

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.
MIL-STD-883 - Test Methods Standard Micrcircuits

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation Production Services (DAPS), Building 4D (DPM–DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

I_P ..... Peak forward pulse current.
$I_{C(ON)}$ ..... On-state collector current.
$I_{C(OFF)}$ ..... Off-state collector current.
R_{IO} ..... Input to output resistance.
V_{IO} ..... Input to output voltage.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1 (similar to T0-99) and 2 (surface mount) herein. With the approval of the qualifying activity, this device is authorized to contain organic material for JANS.

3.4.1 Lead formation and finish for 4N47, 4N47A, 4N48, 4N48A, 4N49, and 4N49A. Lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

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3.4.2 Lead formation and finish for 4N47U, 4N48U, and 4N49U. Terminal pad shall be tungsten co-fired to the ceramic package. Terminal pad finish shall be gold plated or solder dipped. Where a choice of lead finish is desired, it shall be specified in the acquisition document (6.2).

3.4.3 For Eutectic alloyed die of less than 1.5 X 10-5 square inches of surface area. Die shear process control shall be in accordance with the requirements for small area die in method 2019 of MIL-STD-883.

3.4.4 JAN, JANTX, AND JANTXV ONLY. Die shear process control on epoxy mounted die is not required.

3.4.5 Moisture content. The internal moisture content of the device package shall not exceed 10,000 ppm at +100°C.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV level
3	See 4.3.1	See 4.3.1
9	$I_{C(OFF)1}$ and h_{FE}	$I_{C(OFF)1}$ 100 percent read and record
10	See 4.3.2	See 4.3.2
11	I_R, $I_{C(OFF)1}$, h_{FE}, and $I_{C(ON)1}$; $\Delta I_{C(OFF)1}$ = 100 percent of initial value or = 25 nA dc whichever is greater; Δh_{FE} ±20 percent of initial reading.	I_R, $I_{C(OFF)1}$, h_{FE}, and $I_{C(ON)1}$; $\Delta I_{C(OFF)1}$ = 100 percent of initial value or25 nA dc, whichever is greater.
12	See 4.3.3	See 4.3.3
13	Subgroups 2 and 3 of table I herein; $\Delta I_{C(OFF)1}$ = 100 percent of initial value or 25 nA dc, whichever is greater. Δh_{FE} = ±20 percent of initial reading; $\Delta I_{C(ON)1}$ = ±25 percent of initial reading; ΔI_R = 100 percent of initial value or 25 µA dc, whichever is greater.	Subgroup 2 of table I herein; $\Delta I_{C(OFF)1}$ = 100 percent of initial value or 25 nA dc whichever is greater; Δh_{FE} = ±20 percent of initial reading; $\Delta I_{C(ON)1}$ = ±25 percent of initial reading ΔI_R = 100 percent of initial value or 25 µA dc, whichever is greater.

4.3.1 Thermal shock (temperature cycling). All devices shall be subjected to thermal shock (temperature cycling) in accordance with method 1051 of MIL-STD-750, test condition B, except $T_{(min)}$ = -55°C; 10 cycles, time at temperature extremes = 15 minutes minimum.

4.3.1.1 Monitored thermal shock (temperature cycling). One cycle of monitored temperature cycling shall be performed on 100 percent of the devices. This test shall be performed any time after the completion of the thermal shock test specified or it may be the last of the ten thermal cycles. All junctions shall be monitored for electrical continuity. Any discontinuity shall be cause for rejection of the device(s) under test. If 10 percent or more of the number of devices subjected to monitored temperature cycling fail, the entire lot shall be rejected as "TX", "TXV", or "S" types.

4.3.2 High temperature reverse bias (HTRB). All devices shall be subjected to high temperature reverse bias in accordance with method 1039 of MIL-STD-750, test condition A, T_A = +125°C, I_F = 0, V_{CB} = 36 V dc for 48 hours minimum.

4.3.3 Power burn-in conditions. Power burn-in conditions are as follows: See figure 4 for burn-in circuit. V_{CC} = 20 V dc, V_{CE} = 10 ±5 V dc, I_F = 40 mA dc, P_T = 275 ±25 mW at T_A = +25°C ±3°C. No heat sink or forced air cooling directly on the devices shall be permitted.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table III herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1056	Test condition B.
B3	1071	Fine leak: Condition G or H; Gross leak: Condition C
B3	2037	Bond strength: Test condition A. Sample used in decap internal visual design verification may be used in bond strength and die shear. Sample is in units.
B4		High temperature isolation voltage test (see 4.5.2) and as follows: V_{ISO} = 150 V dc, T_A = +125°C, t = 24 hours minimum. n = 22, c = 0, small lot sample size n = 8, c = 0.
B4	1037	V_{CE} = 10 V dc, I_F = 20 mA dc, P_T = 275 ±25 mW at T_A = +25°C ±3°C (see figure 4), t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced air cooling directly on the device shall be permitted.
B5	1027	V_{CE} = 10 V dc, T_A = +100°C ±3°C for 96 hours, P_T = 275 ±25 mW, I_F = 20 mA dc (see figure 4), or adjusted as required by the chosen T_A to give an average lot T_J = +200°C. Marking legibility requirements shall not apply.
B5	2037	(Al-Au die interconnects only), test condition A.
B5	1018	Internal water vapor test, n = 3, c = 0.
B6		Not applicable.

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4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B2	1056	Temperature cycling: Test condition B (25 cycles). Total test time = 72 hours maximum.
B2	1071	Fine leak - condition G or H, gross leak - condition C.
B3		High temperature isolation voltage test conditions: V_{ISO} = 150 V dc (see 4.5.2), T_A = +125°C, t = 24 hours minimum. n = 22, c = 0, small lot sample size = 8, c = 0..
B3	1027	I_F = 20 mA dc, P_T = 275 ± 25 mW at T_A = +25°C ±3°C (see figure 4). No heat sink or forced air cooling directly on the device shall be permitted.
B4	2037	Test condition A.
B5 and B6		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

Subgroup	Method	Condition
C2	1056	Thermal shock (temperature cycling): Test condition A.
C2	2036	Terminal strength, test condition E.
C2	1071	Fine leak - condition G or H; gross leak - condition C.
C3	2016	Nonoperating; 1,500 G, t = 0.5 ms, 5 blows in each orientation: X_1, Y_1, and Y_2.
C3	2056	50 G minimum. Nonoperating.
C3	2006	Nonoperating, 30,000 G, X_1, Y_1, and Y_2, orientations.
C6	1026	I_F = 20 mA dc, P_T = 275 ±25 mW at T_A = +25°C ±3°C (see figure 4). No heat sink or forced air cooling directly on the device shall be permitted.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions of pulse measurements shall be as specified in MIL-STD-750.

4.5.2 High temperature isolation voltage lot verification. This test shall be performed by shorting both the anode and cathode terminals of the LED (light emitting diode) and shorting together the collector, emitter, and base terminals of the transistor. The sample shall be split into two approximately equal groups and the specified isolation voltage (V_{ISO}) applied between the LED terminals (positive) and transistor terminals (negative) of one group. The specified isolation voltage shall be repeated with the LED terminals (negative) and the transistor terminal (positive) for the remaining group. All voltages shall be applied at the specified ambient temperature.

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4.5.3 Input LED tests. These tests shall be performed with the output transistor terminals open.

4.5.4 Output transistor tests. These tests shall be performed with the input LED terminals open.

4.5.5 Isolation and coupling capacitance tests. These tests shall be performed between both input terminals 5 and 7 shorted together and the output terminals (1, 2, and 3, see figure 1) shorted together.

4.6 Internal visual (PRECAP) inspection. Internal visual inspection shall be performed in accordance with methods 2072 and 2073 of MIL-STD-750, and as specified herein. Method 2072 shall be used for inspecting transistor die and completed assembly. Method 2073 shall be used for inspecting light emitting diode.

4.6.1 Gallium arsenide or gallium aluminum arsenide light emitting diode die inspection. The die shall be inspected under 100X magnification.

4.6.2 Silicon phototransistor visual inspection. The die shall be inspected under 100X magnification.

4.6.3 Die position. For applicable constructions, emitter die shall not over-hang the sensor by more than 33 percent of the width of the emitter die.

4.6.4 Die connection. All connections between die assemblies and header posts shall be intact (wires). The connections between the collector post and the header eyelet shall be intact.

4.6.5 Ball support. For applicable constructions, ball bond on cathode contact of light emitter die shall be fully supported underneath with glass coupling preformed and shall not extend to within more than one mil of the edge of the die.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical	2071					
Subgroup 2 2/						
Reverse current leakage	4016	V_R = 2 V dc (see 4.5.3)	I_R		100	µA dc
Forward voltage	4011	I_F = 10 mA dc (see 4.5.3)	V_{F1}	0.8	1.5	V dc
Forward current		V_F = 1.5 V dc (see 4.5.3)	I_F	10.0		mAdc
(Transistor characteristics)						
Collector to emitter breakdown voltage	3011	I_C = 1 mA dc, I_B = 0, I_F = 0 (see 4.5.4)	$V_{(BR)CEO}$	40		V dc
Collector to base breakdown voltage	3001	I_C = 100 µA dc, I_F = 0, I_E = 0 (see 4.5.4)	$V_{(BR)CBO}$	45		V dc
Emitter to base breakdown voltage	3026	I_E = 100 µA dc, I_C = 0, I_F = 0 (see 4.5.4)	$V_{(BR)EBO}$	7		V dc
Off-state collector current (phototransistor mode)	3041	V_{CE} = 20 V dc, I_B = 0, I_F = 0, condition D (see 4.5.4)	$I_{C(OFF)1}$		100	nA dc
Off-state collector current (phototransistor mode)		V_{CB} = 20 V dc, I_F = 0, (see 4.5.4)	$I_{C(OFF)}$		10	nA dc
DC current gain	3076	V_{CE} = 5 V; I_C = 10 mA dc, I_F = 0 (see 4.5.4)	h_{FE}	100		
(Coupler characteristics)						
On-state collector current (phototransistor mode)		V_{CE} = 5 V dc, I_F = 1 mA dc	$I_{C(ON)1}$			
4N47, 4N47A, 4N47U 4N48, 4N48A, 4N48U 4N49, 4N49A, 4N49U				0.5 1.0 2.0	 5.0 10.0	mA dc mA dc mA dc
On-state collector current (phototransistor mode)		V_{CB} = 5 V dc, I_F = 10 mA dc	$I_{CB(ON)}$	30.0		µAdc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued 2/						
Collector to emitter saturation voltage			$V_{CE(SAT)}$			
4N47, 4N47A, 4N47U		I_C = 0.5 mA dc, I_F = 2 mA dc			0.3	V dc
4N48, 4N48A, 4N48U		I_C = 1 mA dc, I_F = 2 mA dc			0.3	V dc
4N49, 4N49A, 4N49U		I_C = 2 mA dc, I_F = 2 mA dc			0.3	V dc
Input to output internal resistance		$\|V_{IN-OUT}\|$ = 1 k V dc (see 4.5.4)	R_{IO}	10^{11}		ohms
Subgroup 3 2/						
High temperature operation		T_A = +100°C				
(Transistor characteristics)						
Off-state collector current (phototransistor mode)	3041	V_{CE} = 20 V dc, I_F = 0, I_B = 0, condition D (see 4.5.4)	$I_{C(OFF)2}$		100	µAdc
(Coupler characteristics)						
On-state collector current (phototransistor mode)	3041	V_{CE} = 5 V dc, I_F = 2 mA dc	$I_{C(ON)2}$			
4N47, 4N47A, 4N47U				0.5		mA dc
4N48, 4N48A, 4N48U				1.0		mA dc
4N49, 4N49A, 4N49U				2.0		mA dc
(LED characteristics)						
Forward voltage	4011	I_F = 10 mA dc (see 4.5.3)	V_{F2}	0.7	1.3	V dc
Low temperature operation		T_A = -55°C				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued 2/						
(Coupler characteristics)						
On-state collector current (phototransistor mode)	3041	V_{CE} = 5 V dc, I_F = 2 mA dc	$I_{C(ON)3}$			
4N47, 4N47A, 4N47U				0.7		mA dc
4N48, 4N48A, 4N48U				1.4		mA dc
4N49, 4N49A, 4N49U				2.8		mA dc
(LED characteristics)						
Forward voltage	4011	I_F = 10 mA dc,(see 4.5.3)	V_{F3}	1.0	1.7	V dc
Subgroup 4 2/						
Input to output capacitance		f = 1 MHz, (see 4.5.5), $\|V_{IN-OUT}\|$ = 0	C_{IO}		5	pF
Rise time (phototransistor mode)		V_{CC} = 10 V dc, I_F = 5 mA dc, R_L = 100 ohms (see figure 3)	t_r			
4N47, 4N47A, 4N47U					20	µs
4N48, 4N48A, 4N48U					20	µs
4N49, 4N49A, 4N49U					25	µs
Fall time (phototransistor mode)		V_{CC} = 10 V dc, I_F = 5 mA dc, R_L = 100 ohms (see figure 3)	t_f		20 20 25	µs µs µs
Rise time (photodiode mode)		V_{CC} = 10 V dc, I_F = 5 mA dc, R_L = 100 ohms (see figure 3)	t_r		3	µs
Fall time (photodiode mode)		V_{CC} = 10 V dc, I_F = 5 mA dc, R_L = 100 ohms (see figure 3)	t_f		3	µs
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ All devices required by the specified sampling plan are subjected to subgroups 2, 3, and 4 combined (JANS only).

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TABLE II. Group E inspections (all quality levels) for qualification only.

Inspections	MIL-STD-750		Quality conformance inspection sample size.
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Thermal shock (temperature cycle)	1051	Test condition B, 500 cycles	
Hermetic seal	1071		
a. Fine leak b. Gross leak		Condition G or H for fine leak Condition C for gross leak	
Electrical measurements		See table III, steps 1, 2, and 3	
Subgroup 2			45 devices, c = 0
High temperature reverse bias	1039	For all devices with organic material.	
Electrical measurements		See table III, steps 1, 2, and 3	

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TABLE III. Groups B, C, and E electrical end-point measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Off-state collector current	3401	V_{CE} = 20 V dc; I_F = 0, condition D (see 4.5.4)	$I_{C(OFF)1}$		100	nA dc
2.	On-state collector current		V_{CE} = 5 V dc, I_F = 1 mA dc	$I_{C(ON)1}$			
	4N47, 4N47A, 4N47U				0.5		mA dc
	4N48, 4N48A, 4N48U				1.0	5.0	mA dc
	4N49, 4N49A, 4N49U				2.0	10.0	mA dc
3.	Collector to emitter saturation voltage			$V_{CE(SAT)}$			
	4N47, 4N47A, 4N47U		I_C = 0.5 mA dc, I_F = 2 mA dc			0.3	V dc
	4N48, 4N48A, 4N48U		I_C = 1 mA dc, I_F = 2 mA dc			0.3	V dc
	4N49, 4N49A, 4N49U		I_C = 2 mA dc, I_F = 2 ma dc			0.3	V dc
4.	DC current gain	3076	V_{CE} = 5 V dc, I_C = 10 mA dc, I_F = 0 (see 4.5.4)	h_{FE}	100		
5.	Reverse current leakage	4016	V = 2 V dc (see 4.5.3)	I_R		100	µA dc
6.	Off-state collector current	3041	V_{CE} = 20 V dc, I_F = 0, condition D, (see 4.5.4)	$\Delta I_{C(OFF)1}$ 4/	±100 percent of initial value or 25 nA dc, whichever is greater.		
7.	On-state collector current		V_{CE} = 5 V dc, I_F =1 mA dc	$\Delta I_{C(ON)1}$ 4/	±25 percent change from initial reading.		
8.	DC current gain	3076	V_{CE} = 5 V dc, I_C = 10 mA dc, I_F = 0 (see 4.5.4)	Δh_{FE} 4/	±25 percent change from initial reading.		
9.	Reverse current leakage	4016	V = 2 V dc, (see 4.5.3)	ΔI_R 4/	±100 percent of initial value or 25 µA dc, whichever is greater.		
10.	Collector to emitter saturation voltage		As in step 3.	$\Delta V_{CE(SAT)}$ 4/	±50 mV dc change from previous measured value.		

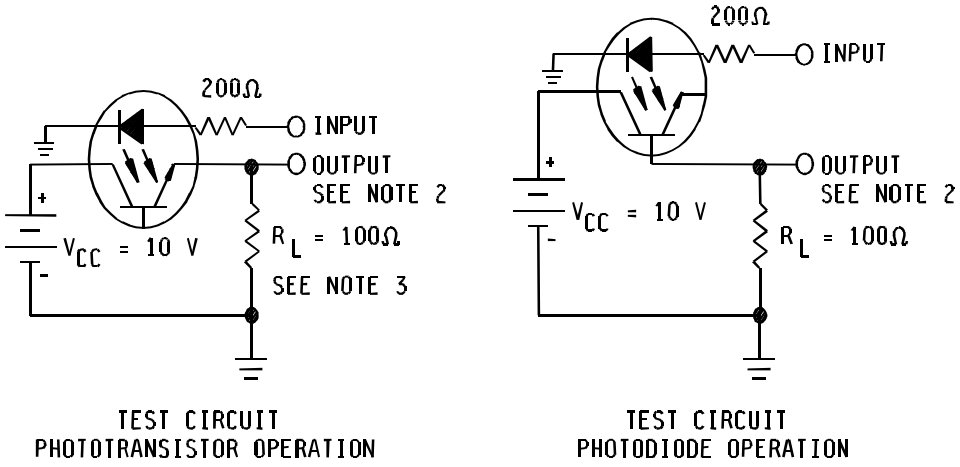
See footnotes at end of table.

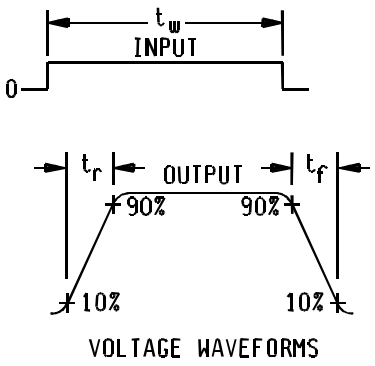
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TABLE III. Groups B, C, and E electrical end-point measurements - Continued. 1/ 2/ 3/

1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table III herein, steps 1, 2, 3, and 5.
   b. Subgroup 4, see table III herein, steps 1, 2, 3, and 5.
   c. Subgroup 5, see table III herein, steps 1, 2, 4, 5, 6, 7, 8, 9, and 10.
2/ The electrical measurements for table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table III herein, steps 1, 2, and 3.
   b. Subgroup 3, see table III herein, steps 1, 2, 5, 6, and 7.
   c. Subgroup 6, see table III herein, steps 6 and 7.
3/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table III herein, steps 1, 2, 3, and 5 (JANS) and steps 1 and 2 (JAN, JANTX, and JANTXV).
   b. Subgroup 3, see table III herein, steps 2 and 3.
4/ Devices which exceed the group A limits for test shall not be acceptable for shipment.

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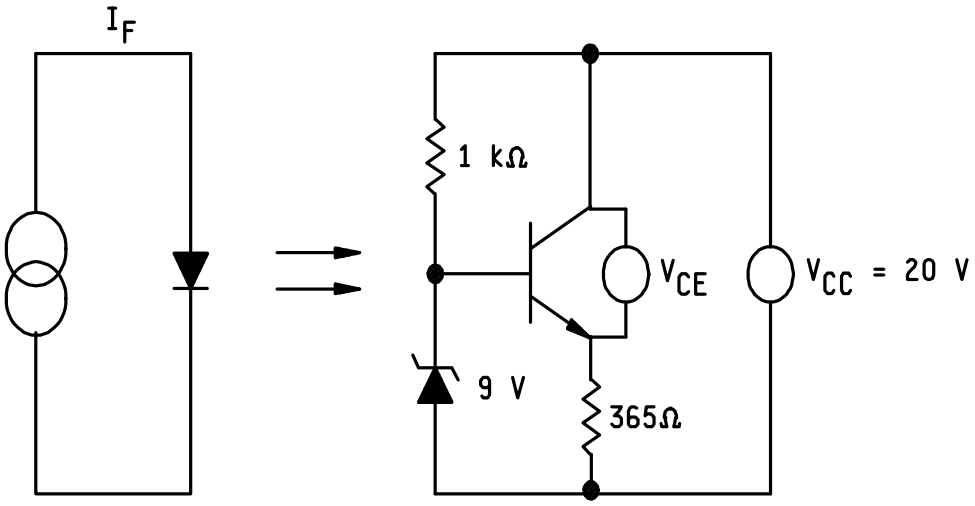


NOTES:

1. The input waveform is supplied by a generator with the following characteristics: Z_{OUT} = 50 Ω, $t_r \leq 15$ ns, duty cycle = 1 percent; t_w = 100 µs.
2. The output waveform is monitored on an oscilloscope with the following characteristics: $t_r \leq 12$ ns, R_{in} = 1 MΩ, C_{in} = 20 pF.
3. Adjust amplitude of input for: I_F = 5 mA dc.

FIGURE 3. Switching times.

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NOTE:
I_F = 20 mA dc for operation life tests.
I_F = 40 mA dc for burn-in of TX, TXV, and S.

FIGURE 4. Operation life test burn-in circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specified issue of individual documents referenced (see 2.2.1).

c. Lead finish (see 3.4.1).

d. Type designation and product assurance level.

e. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturer's List QML-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, P.O. Box 3990 DSCC-VQE, Columbus, OH 43216.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - NW
Air Force - 11
NASA – NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - CC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/548D	2. DOCUMENT DATE 010608

3. **DOCUMENT TITLE**
COUPLER, OPTO ELECTRONIC, SEMICONDUCTOR DEVICE, SOLID STATE TYPES 4N47, 4N47A, 4N48, 4N48A, 4N49, 4N49A, 4N47U, 4N48U, AND 4N49U JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial DSN FAX EMAIL 614-692-0510 850-0510 614-692-6939 alan.barone@dscc.dla.mil
c. ADDRESS Defense Supply Center, Columbus, ATTN: DSCC-VAC P.O. Box 3990 Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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